EmbeddingsBench — findings

Text classification with `jinaai/jina-embeddings-v5-text-small-classification`: classifiers on frozen embeddings, TF-IDF, ensembles, LoRA fine-tuning and a German-native reference model on 18 English and German datasets.

Status: results as of 12 October 2026, 13:00 UTC. Fine-tuning of the remaining datasets and the Swiss imbalance fine-tunes are still running (see the end); the report and the dashboard are updated when they finish. All accuracies are on held-out test splits; every setting (C, k, MLP shape, ensemble weight, fine-tuning checkpoint) was chosen on the validation split only.

Key findings

1. **Logistic regression is the right default on frozen embeddings.** Across 18 datasets a tuned MLP is on average +0.3 points better (median 0.0), an RBF SVM −0.9, kNN −6.5. A 40-trial MLP architecture search mostly settles on shallow networks (depth 0–1 on 13 of 18 datasets) whose gain over LogReg is within noise.
2. **The cheapest real gain is an ensemble with TF-IDF.** Averaging LogReg probabilities with a word-n-gram TF-IDF SVM beats LogReg by more than 0.5 points on 13 of 18 datasets, +3.5 points on average, with no GPU training. It matches or beats LoRA fine-tuning on 6 of the 10 datasets where both were run.
3. **The best jina-based classifier is fine-tuned jina + TF-IDF**: +3.9 points over frozen LogReg on average over 9 datasets, ahead of every classifier without GPU training on 7 of them (Yahoo! 72.9 %, arXiv 88.6 %, 20 Newsgroups 75.6 %, 10kGNAD 89.9 %, CLINC150 95.9 %). Fine-tuning and lexical features are complementary.
4. **After fine-tuning, the classifier on top no longer matters.** LogReg, RBF SVM and a tuned MLP on the fine-tuned encoder's embeddings land within 2 points (mostly within 1) of its jointly trained linear head (the MLP search chose a linear model on every dataset). kNN trails LogReg by up to 20 points on frozen embeddings, but only by 0–6 points after fine-tuning, because fine-tuning pulls each class together.
5. **For German, a German-native encoder fine-tuned in full is stronger.** `gbert-large` beat LoRA-fine-tuned jina on all three German datasets: 10kGNAD 91.5 vs 88.6 % (89.9 % with TF-IDF), Amazon reviews 64.7 vs 63.6 %, MASSIVE 88.0 vs 87.0 %. It needed 21–42 min of training and under 4 GiB of GPU memory per dataset (8-bit AdamW).
6. **Frozen embeddings struggle with many fine-grained classes.** With 342 food categories (Open Food Facts DE) LogReg reaches 44.9 % while TF-IDF reaches 57.8 % and the ensemble with character n-grams 60.6 %. For 408 patent classes (HUPD): 50.2 / 52.3 / 54.3 %. Intent datasets with 60–150 classes are the exception (87–95 %), because their classes differ in meaning, not in vocabulary.
7. **Long, technical or legal texts favour lexical features.** On Swiss court decisions, arXiv and LEDGAR, TF-IDF alone beats LogReg on frozen embeddings by 3–7 points and is within 1.5 points of the best method.
8. **Class weighting buys rare-class recall; the price grows with the number of classes.** Tail recall (the third of classes with the fewest training examples): MTOP 27.9 → 36.8 % at equal accuracy, Swiss decisions 79.4 → 87.7 % (−1.5 points accuracy), Open Food Facts 36.4 → 46.8 % (−2.4), HUPD 12.6 → 29.8 % (−5.9), GermEval 2.2 → 17.8 % (−7.9). Where a large catch-all or a few small classes dominate, it is free: Medical abstracts balanced accuracy 67.3 → 73.6 %, Emotion 86.2 → 90.0 %, both at unchanged accuracy.
9. **Zero-shot (label-name similarity) only works when label names are descriptive**: 91 % on emotions, 82 % on banking intents, under 10 % on legal areas, patents and product categories.
10. **2,048-token input only matters when most texts are long** (Swiss decisions +2.2, 10kGNAD +0.9, arXiv +0.5 points vs 512 tokens); for fine-tuning, head + tail truncation helped more than

1,024 tokens (10kGNAD 89.6 vs 88.8 %). **Matryoshka truncation to 512 dimensions costs 1.1 points on average** (at most 3.1), 256 dimensions 2.8 points.

11. **Many top results are statistically tied.** The 95 % bootstrap interval of the best method's accuracy is ±0.4 to ±1.9 points; on 11 of 18 datasets at least one other method is statistically tied with the best. Differences below about 1 point should not drive the choice of method.
12. **LogReg's confidence can route texts to manual review.** Its calibration error is mostly below 7 % (up to 16 % on LEDGAR). At a 95 % accuracy target, 93–100 % of texts can be classified automatically on Banking77, CLINC150, MTOP, Emotion and DBpedia, 86 % on Swiss decisions, 43 % on MASSIVE, but only 0–7 % on sentiment, offensive language, product categories and patents. The TF-IDF ensembles' scores are not calibrated (ECE up to 68 %) and need recalibration before thresholding.

Setup

- **Encoder:** `jinaai/jina-embeddings-v5-text-small-classification` (revision `4447914`), last-token pooling, L2-normalised 1,024-d embeddings, prompt `Document:`, up to 2,048 tokens.
- **Classifiers on frozen embeddings:** zero-shot (cosine similarity to label-name embeddings), kNN (cosine, k tuned), logistic regression (C tuned), RBF SVM, MLP (1–2 GELU layers) and a 40-trial MLP architecture search.
- **Text baselines:** TF-IDF (word 1–2-grams, sublinear tf) with a linear SVM; character 2–5-gram TF-IDF.
- **Ensembles:** weighted average of LogReg probabilities and softmaxed TF-IDF scores; weight and temperature chosen on validation.
- **Fine-tuning:** LoRA rank 16 on all attention and MLP projections, linear head initialised from the frozen LogReg, 3 epochs, 512 tokens (variants: 1,024 tokens, head + tail truncation, class-weighted and logit-adjusted loss), plus an ensemble with TF-IDF and the full classifier sweep on the fine-tuned embeddings. `deepset/gbert-large` fully fine-tuned as a German reference (8-bit AdamW, effective batch 16, 3 epochs).
- **Metrics:** accuracy, balanced accuracy, macro-F1, MCC, Cohen's κ, top-3 accuracy and tail recall (recall on the third of classes with the fewest training examples).
- **Hardware:** one RTX 3060 (12 GB), shared with other services.
- **Method categories** (as grouped in the dashboard): frozen jina + classifier · frozen jina with imbalance schemes · text only (TF-IDF) · ensembles of frozen embeddings and TF-IDF · fine-tuned jina (LoRA) · classifiers on fine-tuned jina embeddings · other model (gbert-large).

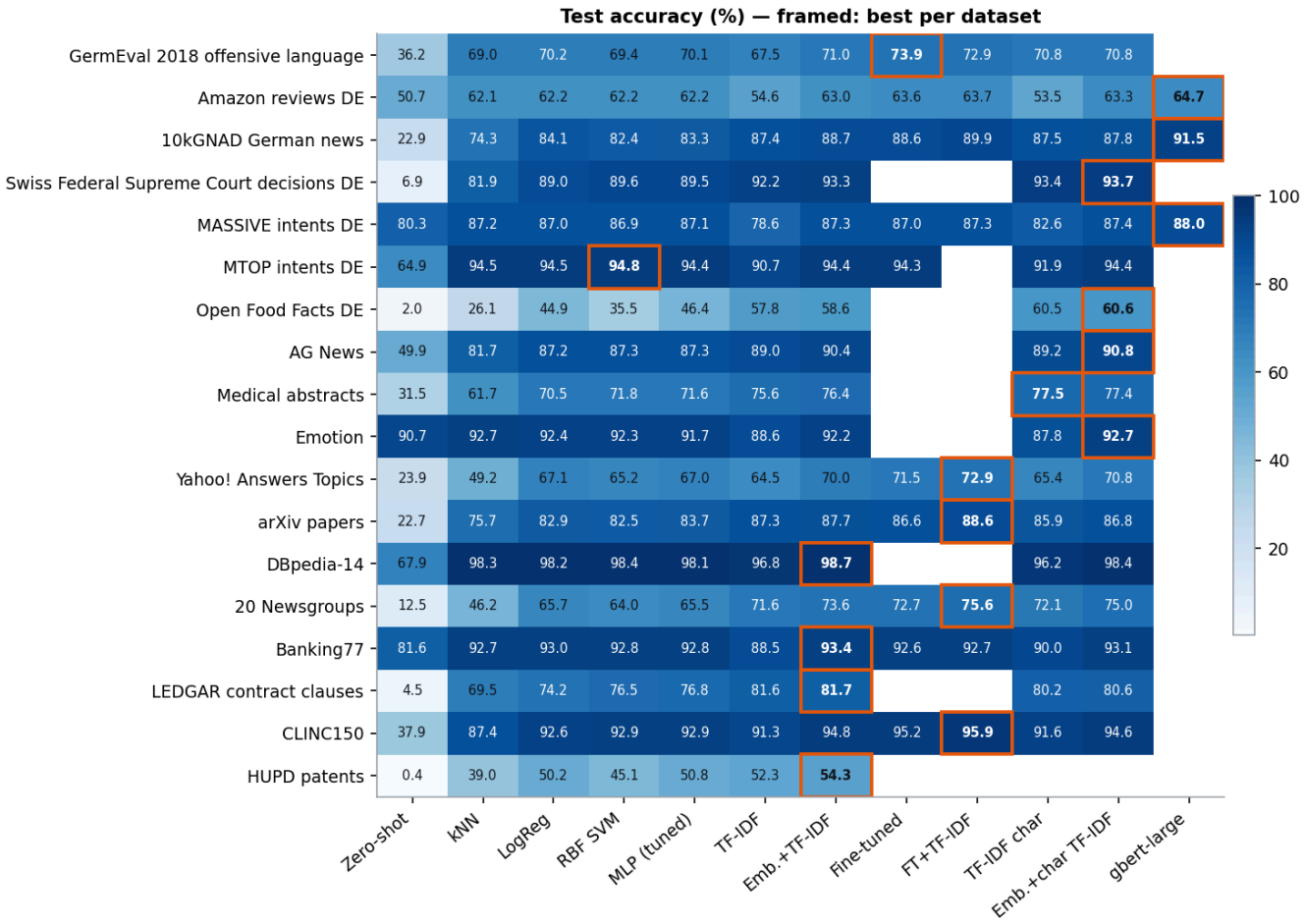
Datasets

Dataset	Lang	Domain	Classes	Train / val / test	Avg tokens	Train imbalance
GermEval 2018 offensive language	DE	tweets (offensive language)	4	4,258 / 751 / 3,398	45	47:1
Amazon reviews DE (stars)	DE	product reviews (sentiment)	5	10,000 / 5,000 / 4,999	65	1:1
10kGNAD German news	DE	news	9	7,860 / 1,384 / 1,027	732	3:1
Swiss Federal Supreme Court decisions DE (legal area)	DE	court decisions	47	25,737 / 3,002 / 3,002	3413	19:1
MASSIVE intents DE	DE	assistant intents	60	11,514 / 1,943 / 2,845	12	202:1
MTOP intents DE	DE	assistant intents (long tail)	103	11,072 / 1,956 / 1,756	14	1175:1

Dataset	Lang	Domain	Classes	Train / val / test	Avg tokens	Train imbalance
Open Food Facts DE (product categories)	DE	food products	342	69,366 / 8,790 / 8,790	18	12:1
AG News	EN	news	4	10,000 / 2,000 / 7,599	54	1:1
Medical abstracts	EN	medical	5	9,817 / 1,221 / 1,878	274	3:1
Emotion (tweets)	EN	emotion (tweets)	6	16,000 / 1,995 / 1,986	22	9:1
Yahoo! Answers Topics	EN	community Q&A	10	10,000 / 2,000 / 5,000	127	1:1
arXiv papers (11 categories)	EN	scientific papers	11	8,800 / 2,446 / 2,441	4300	1:1
DBpedia-14	EN	encyclopedia	14	7,000 / 1,400 / 2,800	79	1:1
20 Newsgroups	EN	forum posts	20	9,361 / 1,644 / 7,299	275	2:1
Banking77	EN	banking intents	77	8,497 / 1,496 / 3,069	15	5:1
LEDGAR contract clauses (LexGLUE)	EN	legal contracts	100	19,092 / 9,914 / 9,855	144	9:1
CLINC150 (in-scope)	EN	assistant intents	150	15,000 / 2,997 / 4,498	11	1:1
HUPD patents (IPC subclass)	EN	patents (IPC subclasses, long tail)	408	98,195 / 18,934 / 18,901	151	50:1

Splits are fixed and stratified where the source has none; large sources are capped per class (see `bench/data.py`). Train imbalance is the ratio of the largest to the smallest class in the training split.

Overview

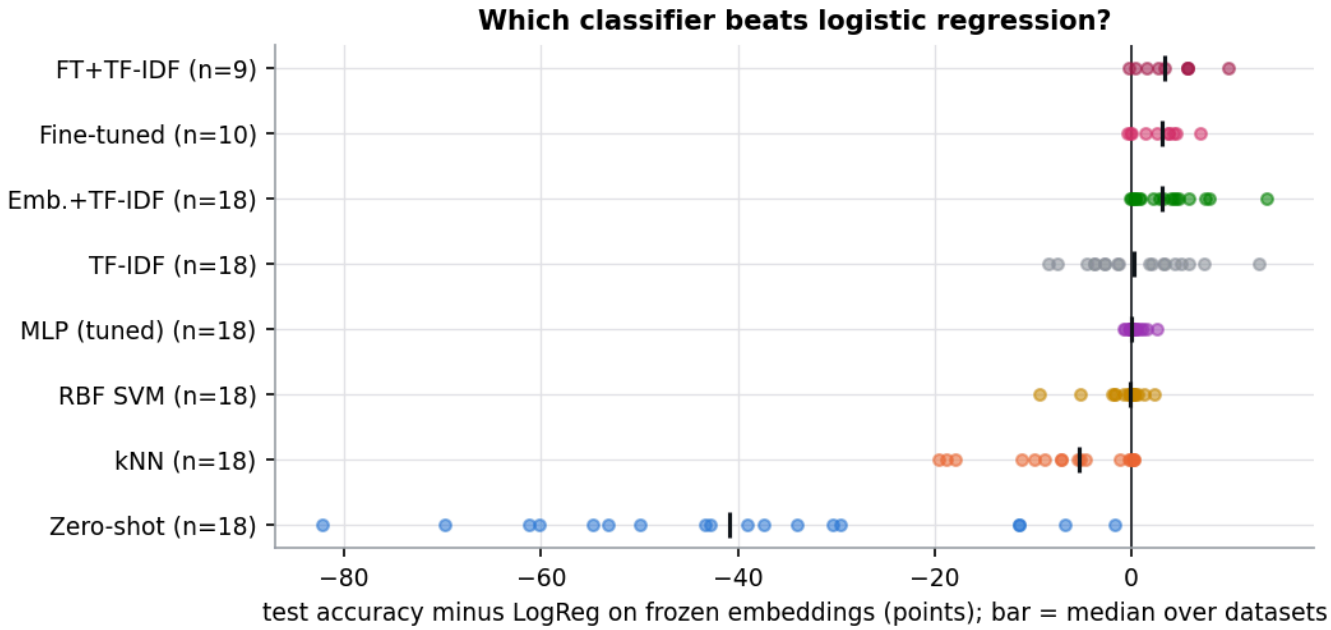


Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
GermEval 2018 offensive language	4	36.2	69.0	70.2	70.3	67.5	71.0	**73.9** (Fine-tuned jina)	47.4
Amazon reviews DE	5	50.7	62.1	62.2	62.0	54.6	63.0	**64.7** (gbert-large)	64.6
10kGNAD German news	9	22.9	74.3	84.1	83.7	87.4	88.7	**91.5** (gbert-large)	91.5
Swiss Federal Supreme Court decisions DE	47	6.9	81.9	89.0	89.0	92.2	93.3	**93.7** (Emb.+char TF-IDF)	89.4
MASSIVE intents DE	60	80.3	87.2	87.0	87.0	78.6	87.3	**88.0** (gbert-large)	84.1
MTOP intents DE	103	64.9	94.5	94.5	94.6	90.7	94.4	**94.8** (RBF SVM)	83.2
Open Food Facts DE	342	2.0	26.1	44.9	45.4	57.8	58.6	**60.6** (Emb.+char TF-IDF)	56.9
AG News	4	49.9	81.7	87.2	86.9	89.0	90.4	**90.8** (Emb.+char TF-IDF)	90.7
Medical abstracts	5	31.5	61.7	70.5	72.0	75.6	76.4	**77.5** (TF-IDF char)	77.2
Emotion	6	90.7	92.7	92.4	92.3	88.6	92.2	**92.7** (Emb.+char TF-IDF)	88.1
Yahoo! Answers Topics	10	23.9	49.2	67.1	67.6	64.5	70.0	**72.9** (Fine-tuned jina + TF-IDF)	72.5
arXiv papers	11	22.7	75.7	82.9	82.8	87.3	87.7	**88.6** (Fine-tuned jina + TF-IDF)	87.9

Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
DBpedia-14	14	67.9	98.3	98.2	98.2	96.8	98.7	**98.7** (Emb.+TF-IDF)	98.7
20 Newsgroups	20	12.5	46.2	65.7	65.5	71.6	73.6	**75.6** (Fine-tuned jina + TF-IDF)	74.0
Banking77	77	81.6	92.7	93.0	92.1	88.5	93.4	**93.4** (Emb.+TF-IDF)	93.3
LEDGAR contract clauses	100	4.5	69.5	74.2	75.1	81.6	81.7	**81.7** (Emb.+TF-IDF)	77.6
CLINC150	150	37.9	87.4	92.6	92.3	91.3	94.8	**95.9** (Fine-tuned jina + TF-IDF)	95.9
HUPD patents	408	0.4	39.0	50.2	50.4	52.3	54.3	**54.3** (Emb.+TF-IDF)	38.7

The best method per dataset is picked by test accuracy for this overview; each method's own settings were chosen on validation. On GermEval 2018 and the long-tail sets (MTOP, HUPD), macro-F1 is far below accuracy: the majority classes dominate accuracy.

Classifiers on top of the embeddings

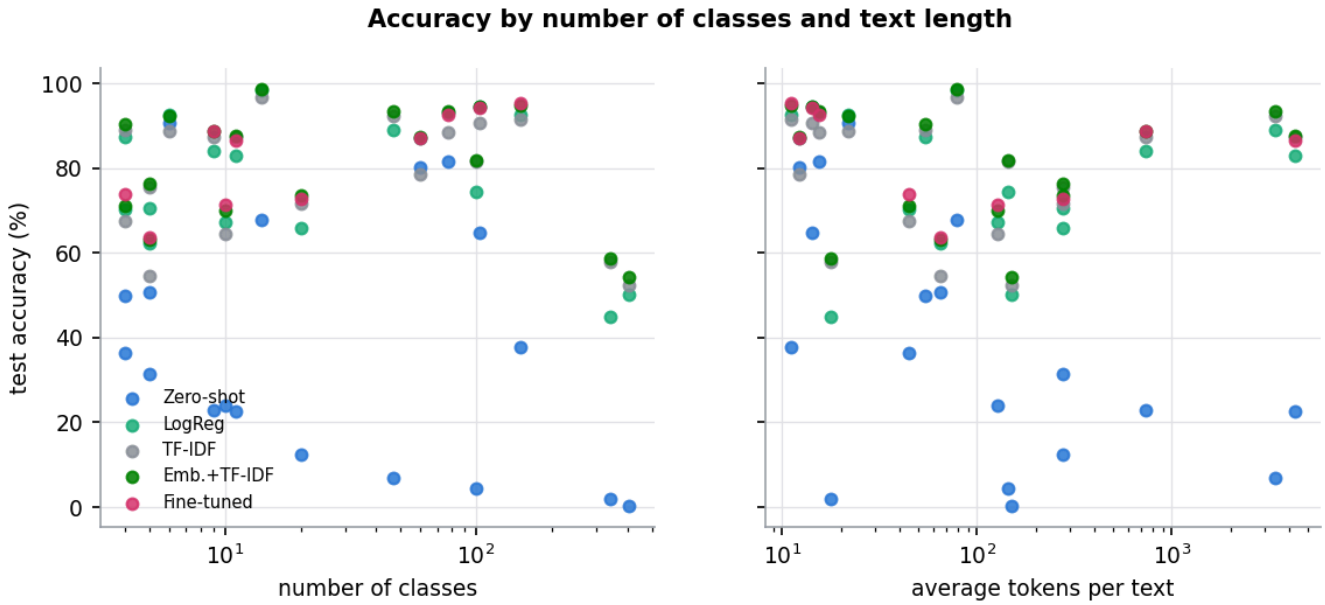


Test accuracy difference to LogReg on frozen embeddings, in points:

Classifier	Mean Δ (pts)	Median Δ (pts)	Better by > 0.5 pts	Datasets
FT+TF-IDF	+3.9	+3.3	7	9
Emb.+TF-IDF	+3.5	+3.0	13	18
Fine-tuned	+2.7	+3.1	7	10
TF-IDF	+0.5	+0.3	9	18
MLP (tuned)	+0.3	+0.0	5	18
RBF SVM	-0.9	-0.1	3	18
kNN	-6.5	-5.3	0	18
Zero-shot	-39.9	-40.9	0	18

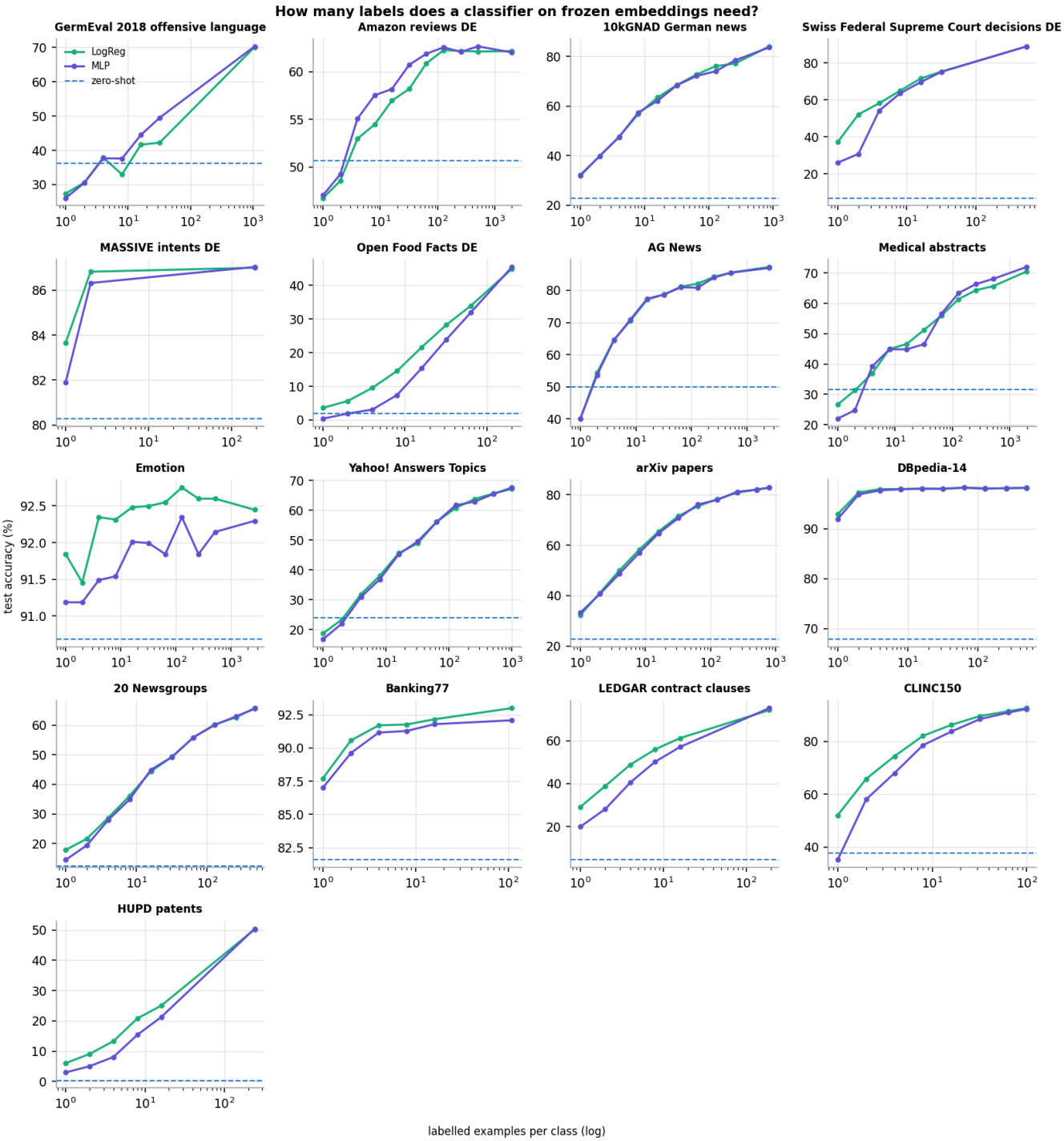
- **LogReg vs MLP:** the embeddings are already linearly separable where they separate at all. A tuned MLP gains more than 0.5 points on only 5 of 18 datasets (most on LEDGAR, +2.6) and small amounts on the many-class sets (Open Food Facts +1.5, HUPD +0.6, Swiss decisions +0.5). The search chose depth 0 on 4 datasets, depth 1 on 9, depth 2–3 on 5.
- **RBF SVM:** never clearly better than LogReg, slower to train (a 30k subsample was needed above 30k training texts) and on Open Food Facts 9 points worse.
- **kNN:** competitive on intents and emotions (within 0.5 points of LogReg on Banking77, MTOP, MASSIVE, Emotion), weak on topical and many-class data (−18 points on Yahoo!, −19 on Open Food Facts).
- **Zero-shot:** useful as a no-label starting point only where label names describe the class well.

Number of classes and text length



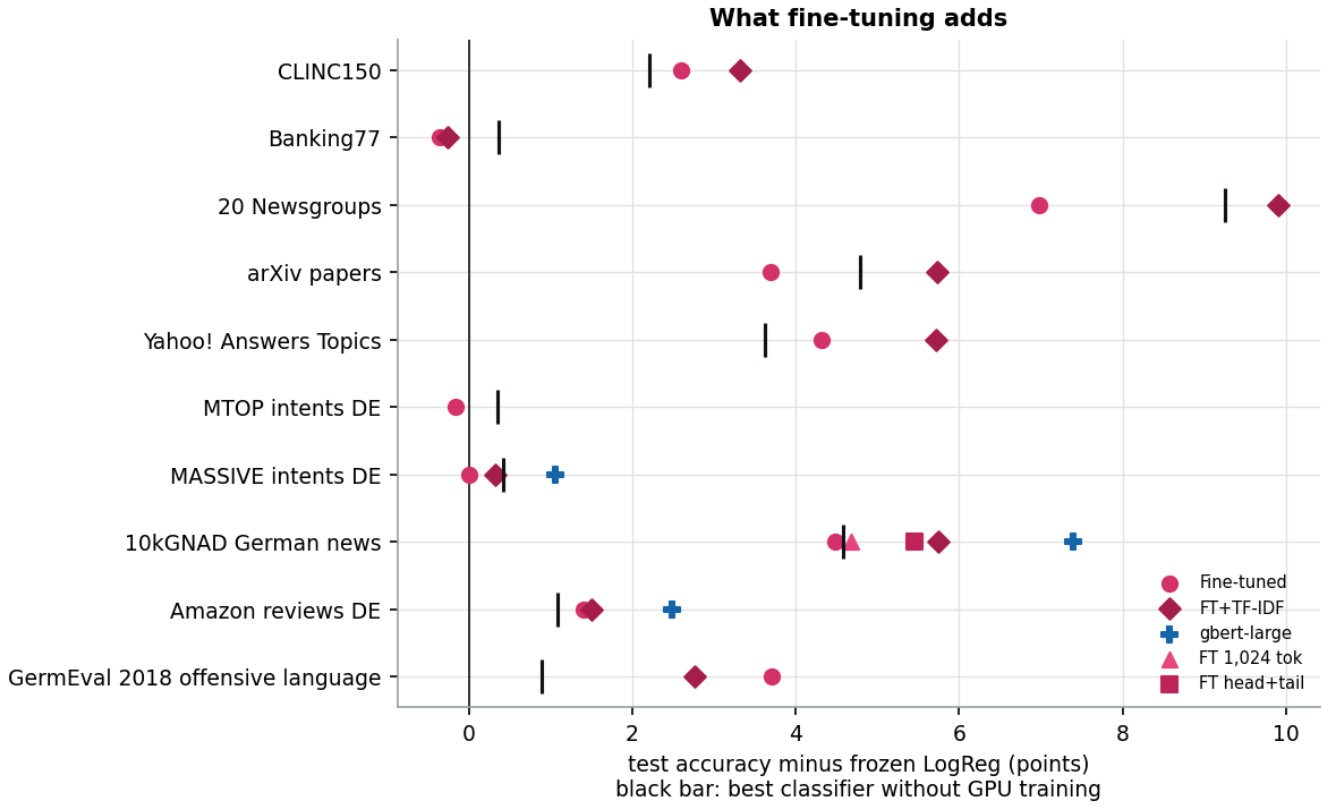
Accuracy on frozen embeddings drops sharply above about 150 classes, while TF-IDF degrades less: a single 1,024-d embedding appears to blur fine distinctions between neighbouring product or patent categories that specific words and word parts still separate. Text length alone is not the driver: short intent texts are classified well and short product names badly.

How many labels are needed



With 8–16 examples per class, LogReg on frozen embeddings already reaches most of its full-data accuracy on intent, emotion and encyclopedia data (Banking77 ~92 % at 16 per class, DBpedia ~98 % at 4). Topical and many-class datasets keep improving up to the full training set; there, more labels pay off more than a different classifier. LogReg is mostly ahead of the MLP at small sample sizes.

Fine-tuning

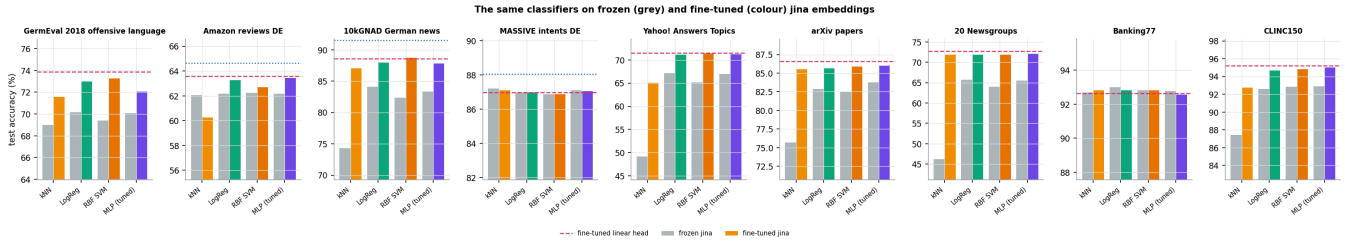


Dataset	Frozen LogReg	Fine-tuned	Best without GPU training	FT + TF-IDF	gbert-large	Train minutes
GermEval 2018 offensive language	70.2	73.9	71.0	72.9	–	10
Amazon reviews DE	62.2	63.6	63.3	63.7	64.7	37
10kGNAD German news	84.1	88.6	88.7	89.9	91.5	76
MASSIVE intents DE	87.0	87.0	87.4	87.3	88.0	26
MTOP intents DE	94.5	94.3	94.8	–	–	25
Yahoo! Answers Topics	67.1	71.5	70.8	72.9	–	39
arXiv papers	82.9	86.6	87.7	88.6	–	150
20 Newsgroups	65.7	72.7	75.0	75.6	–	46
Banking77	93.0	92.6	93.4	92.7	–	19
CLINC150	92.6	95.2	94.8	95.9	–	34

- Fine-tuning pays off where classes are topical or subtle and the frozen embedding is not tuned to the distinction (Yahoo! +4.3, 20NG +7.0, 10kGNAD +4.5, GermEval +3.7, CLINC150 +2.6 points over frozen LogReg). It brings nothing on Banking77 (−0.4), MTOP (−0.2) and MASSIVE (0.0); on MASSIVE the best checkpoint was step 0, i.e. the untouched encoder.
- Fine-tuned jina alone beats the best classifier without GPU training on 4 of 10 datasets (Yahoo!, CLINC150, Amazon, GermEval); **combined with TF-IDF it is ahead on 7 of 9** (−0.6 to +2.1 points).
- 10kGNAD variants: head + tail truncation (first 384 + last 128 tokens) 89.6 %, 1,024 tokens 88.8 %, 512 tokens 88.6 %.
- `gbert-large` (335M parameters, all weights trained) is the best model on every German dataset where it ran. It is a different encoder, so its predictions cannot reuse stored jina embeddings.

Classifiers on fine-tuned embeddings

Does it pay to run the classifier sweep again after fine-tuning? The fine-tuned encoder's pooled, normalised embeddings were computed for train / val / test, and the same classifiers with the same grids were fitted on them.



Dataset	Frozen kNN	Frozen LogReg	Frozen RBF SVM	Frozen MLP (tuned)	FT-emb kNN	FT-emb LogReg	FT-emb RBF SVM	FT-emb MLP (tuned)	Fine-tuned	gbert-large
GermEval 2018 offensive language	69.0	70.2	69.4	70.1	71.6	73.0	73.2	72.1	73.9	–
Amazon reviews DE	62.1	62.2	62.2	62.2	60.3	63.3	62.7	63.5	63.6	64.7
10kGNAD German news	74.3	84.1	82.4	83.3	87.0	87.9	88.7	87.8	88.6	91.5
MASSIVE intents DE	87.2	87.0	86.9	87.1	87.1	87.0	86.9	87.1	87.0	88.0
Yahoo! Answers Topics	49.2	67.1	65.2	67.0	65.1	71.1	71.4	71.2	71.5	–
arXiv papers	75.7	82.9	82.5	83.7	85.5	85.6	85.9	86.0	86.6	–
20 Newsgroups	46.2	65.7	64.0	65.5	71.8	71.8	71.9	72.1	72.7	–
Banking77	92.7	93.0	92.8	92.8	92.8	92.8	92.8	92.6	92.6	–
CLINC150	87.4	92.6	92.9	92.9	92.8	94.7	94.8	95.0	95.2	–

- LogReg, RBF SVM and MLP on the fine-tuned embeddings end up within 2 points (mostly within 1) of the fine-tuned linear head; none is consistently better. kNN is level with LogReg on MASSIVE, 20NG, arXiv and Banking77, 1–2 points lower on 10kGNAD, GermEval and CLINC150, and 3–6 points lower on Amazon and Yahoo!. The MLP search picked a linear model on every dataset. **Keep the jointly trained linear head.**
- kNN profits most from fine-tuning (+13 to +26 points over frozen kNN on 10kGNAD, Yahoo! and 20NG). It becomes a reasonable choice when classes change often, since new labelled examples can be added without retraining. On sentiment (Amazon) it stays weakest.
- Caveat: the encoder saw the training texts, so the training embeddings are tighter than unseen ones; the test texts were never seen.

Class imbalance



Plain vs class-weighted LogReg on every dataset (%; tail recall: recall on the third of classes with the fewest training examples):

Dataset	Train imbalance	LogReg acc.	weighted	LogReg bal. acc.	weighted	LogReg tail recall	weighted
GermEval 2018 offensive language	47:1	70.2	62.3	35.5	45.2	2.2	17.8
Amazon reviews DE	1:1	62.2	62.2	62.2	62.2	74.8	74.8
10kGNAD German news	3:1	84.1	82.6	82.5	82.3	73.9	79.3
Swiss Federal Supreme Court decisions DE	19:1	89.0	87.5	83.9	86.1	79.4	87.7
MASSIVE intents DE	202:1	87.0	86.7	84.0	84.7	72.0	74.7
MTOP intents DE	1175:1	94.5	94.5	81.8	86.0	27.9	36.8
Open Food Facts DE	12:1	44.9	42.5	42.4	46.2	36.4	46.8
AG News	1:1	87.2	87.2	87.2	87.2	82.1	82.1
Medical abstracts	3:1	70.5	70.5	67.3	73.6	53.0	81.9
Emotion	9:1	92.4	92.3	86.2	90.0	70.1	85.8
Yahoo! Answers Topics	1:1	67.1	67.1	67.1	67.1	59.1	59.1
arXiv papers	1:1	82.9	82.9	82.3	82.3	72.8	72.8
DBpedia-14	1:1	98.2	98.2	98.2	98.2	98.5	98.5
20 Newsgroups	2:1	65.7	65.5	64.3	64.4	49.2	51.3
Banking77	5:1	93.0	93.0	93.0	93.0	94.0	94.0
LEDGAR contract clauses	9:1	74.2	74.1	71.5	72.7	67.0	69.8
CLINC150	1:1	92.6	92.6	92.6	92.6	93.3	93.3
HUPD patents	50:1	50.2	44.3	33.1	40.5	12.6	29.8

All schemes on the five most imbalanced or many-class datasets (accuracy / balanced accuracy / macro-F1 / tail recall, %); the other datasets are in `report/tables.md`:

Swiss Federal Supreme Court decisions DE (legal area)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	89.0	83.9	81.4	79.4
LogReg weighted	87.5	86.1	79.6	87.7
Logit-adjusted	89.0	83.9	81.4	79.4
MLP (tuned)	89.5	83.6	80.5	80.4
MLP weighted	86.7	84.4	78.1	88.2
MLP balanced	87.8	85.1	80.3	84.9
TF-IDF	92.2	86.6	86.0	83.9
Emb.+TF-IDF	93.3	88.0	87.5	85.3

MTOP intents DE

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	94.5	81.8	80.6	27.9
LogReg weighted	94.5	86.0	82.0	36.8
Logit-adjusted	93.7	87.0	78.4	40.2
MLP (tuned)	94.4	83.9	81.9	33.8
MLP weighted	93.8	85.5	79.0	36.8
MLP balanced	94.4	86.4	83.9	37.7
TF-IDF	90.7	69.8	67.1	16.2

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
Emb.+TF-IDF	94.4	81.7	80.5	27.9

Open Food Facts DE (product categories)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	44.9	42.4	42.8	36.4
LogReg weighted	42.5	46.2	42.3	46.8
Logit-adjusted	44.9	42.4	42.8	36.4
MLP (tuned)	46.4	43.4	43.7	38.3
MLP weighted	41.6	47.5	41.9	51.4
MLP balanced	42.3	47.7	43.0	50.7
TF-IDF	57.8	56.5	55.1	53.4
Emb.+TF-IDF	58.6	56.9	55.7	53.9

LEDGAR contract clauses (LexGLUE)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	74.2	71.5	68.3	67.0
LogReg weighted	74.1	72.7	68.8	69.8
Logit-adjusted	74.2	71.5	68.3	67.0
MLP (tuned)	76.8	72.6	69.7	68.1
MLP weighted	76.4	72.7	69.3	69.2
MLP balanced	74.5	71.8	68.4	68.8
TF-IDF	81.6	80.0	77.5	77.4
Emb.+TF-IDF	81.7	80.1	77.6	77.4

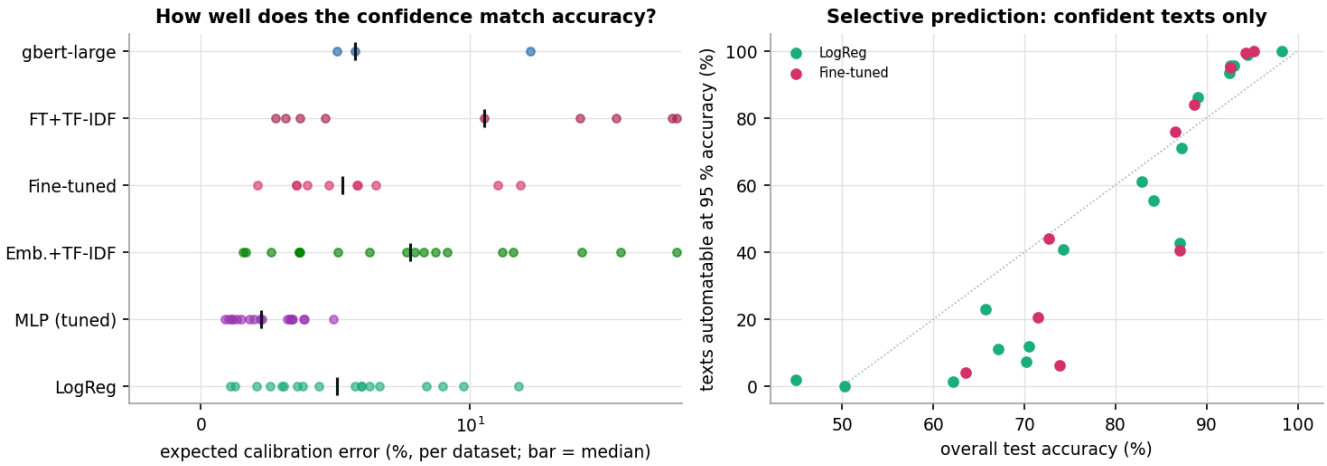
HUPD patents (IPC subclass)

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	50.2	33.1	32.9	12.6
LogReg weighted	44.3	40.5	33.7	29.8
Logit-adjusted	48.8	38.4	34.3	22.2
MLP (tuned)	50.8	38.7	36.5	24.6
MLP weighted	40.7	42.8	31.5	39.3
MLP balanced	42.7	42.9	33.7	34.9
TF-IDF	52.3	39.6	38.3	25.3
Emb.+TF-IDF	54.3	39.2	38.7	22.7

- **Class weighting is free on balanced data** (identical results on Amazon, Yahoo!, AG News, CLINC150, DBpedia, arXiv) and **free and useful where a catch-all class absorbs predictions**: on Medical abstracts recall on the two smallest disease classes rises by 29 and 19 points while the large catch-all class "general pathological conditions" drops by 22; accuracy is unchanged and balanced accuracy +6.3. On GermEval (47:1) it lifts the rarest class but costs 7.9 points of accuracy.
- **Class-weighted LogReg** raises tail recall on every imbalanced dataset: LEDGAR +2.8, MTOP +8.9, Swiss decisions +8.3, Open Food Facts +10.4, HUPD +17.2 points. The cost in accuracy grows with the number of classes and the imbalance (0 on MTOP and LEDGAR, −1.5 Swiss, −2.4 Open Food Facts, −5.9 HUPD).

- **Logit adjustment** (subtracting τ · log prior at prediction time) is free to apply after training. τ is chosen on validation: τ = 0 (no change) on LEDGAR, Swiss and Open Food Facts; on MTOP and HUPD it improves balanced accuracy (+5.2, +5.3 points); on HUPD at a much smaller accuracy cost than weighting (−1.4 vs −5.9 points).
- **MLP with weighted loss or balanced batches** gives the highest tail recall of the embedding classifiers on Open Food Facts and HUPD, again at the cost of accuracy.
- TF-IDF and the ensemble lead on rare classes on LEDGAR (77.4 %) and Open Food Facts (53.4 / 53.9 %); on Swiss decisions the class-weighted embedding classifiers are ahead (87.7–88.2 vs 83.9 %), and on MTOP (1175:1, short texts) TF-IDF is weakest: 16 % tail recall.
- **Weighted and logit-adjusted fine-tuning on MTOP**: accuracy 94.3 (plain) → 94.7 (weighted) / 94.8 (logit-adjusted), macro-F1 79.8 → 82.3 / 81.9; i.e. small gains on both. The Swiss decisions runs are being repeated after an out-of-memory failure.

Reliability



Significance. 1,000 bootstrap resamples of each test set give a 95 % interval for every method's accuracy and, paired over the same resamples, for its difference to the best method. A method counts as *tied with the best* when that difference interval includes 0. This is also marked in the dashboard (thin frame, interval on hover).

Dataset	Best (95 % interval)	Half-width (pts)	Statistically tied with the best
GermEval 2018 offensive language	Fine-tuned 73.9 [72.5–75.4]	±1.4	–
Amazon reviews DE	gbert-large 64.7 [63.2–66.0]	±1.4	Fine-tuned, Emb.+char TF-IDF, FT+TF-IDF
10kGNAD German news	gbert-large 91.5 [89.9–93.2]	±1.7	–
Swiss Federal Supreme Court decisions DE	Emb.+char TF-IDF 93.7 [92.9–94.6]	±0.8	Emb.+TF-IDF, TF-IDF char
MASSIVE intents DE	gbert-large 88.0 [86.9–89.2]	±1.2	kNN, Emb.+TF-IDF, MLP (tuned), Emb.+char TF-IDF, FT+TF-IDF
MTOP intents DE	RBF SVM 94.8 [93.8–95.8]	±1.0	kNN, LogReg, MLP, Emb.+TF-IDF, Emb.+char TF-IDF, MLP (tuned), LogReg weighted, MLP balanced, Fine-tuned, FT weighted
Open Food Facts DE	Emb.+char TF-IDF 60.6 [59.7–61.7]	±1.0	TF-IDF char
AG News	Emb.+char TF-IDF 90.8 [90.1–91.4]	±0.6	Emb.+TF-IDF

Dataset	Best (95 % interval)	Half-width (pts)	Statistically tied with the best
Medical abstracts	TF-IDF char 77.5 [75.6–79.4]	±1.9	Emb.+TF-IDF, Emb.+char TF-IDF
Emotion	Emb.+char TF-IDF 92.7 [91.5–93.9]	±1.2	kNN, LogReg, MLP, RBF SVM, Emb.+TF-IDF, LogReg weighted, Logit-adjusted, MLP weighted, MLP balanced
Yahoo! Answers Topics	FT+TF-IDF 72.9 [71.7–74.0]	±1.2	–
arXiv papers	FT+TF-IDF 88.6 [87.4–89.9]	±1.3	–
DBpedia-14	Emb.+TF-IDF 98.7 [98.2–99.1]	±0.4	–
20 Newsgroups	FT+TF-IDF 75.6 [74.6–76.6]	±1.0	Emb.+char TF-IDF
Banking77	Emb.+TF-IDF 93.4 [92.5–94.3]	±0.9	Emb.+char TF-IDF, MLP balanced
LEDGAR contract clauses	Emb.+TF-IDF 81.7 [81.0–82.5]	±0.7	TF-IDF
CLINC150	FT+TF-IDF 95.9 [95.3–96.4]	±0.6	–
HUPD patents	Emb.+TF-IDF 54.3 [53.5–55.0]	±0.7	–

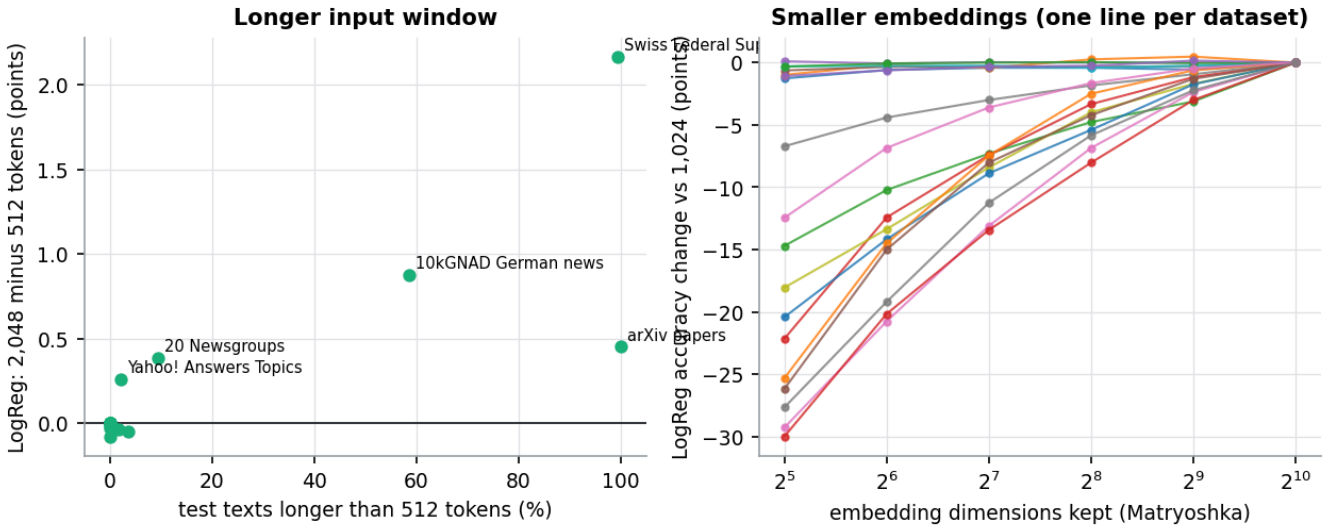
Calibration and selective prediction. ECE is the gap between confidence and accuracy, averaged over 15 confidence bins. *Automatable at 95 %* is the largest share of test texts that can be classified automatically, taking the most confident first, while keeping 95 % accuracy; the rest would go to manual review. The confidence thresholds are in `results.json` (`reliability`) and in the dashboard.

Dataset	Classifier	Accuracy	ECE	Accuracy on the 80 % most confident	Automatable at 95 % accuracy	at 99 %
GermEval 2018 offensive language	LogReg	70.2	3.1	76.9	7	0
GermEval 2018 offensive language	Fine-tuned	73.9	16.2	80.3	6	0
Amazon reviews DE	LogReg	62.2	4.4	66.6	1	0
Amazon reviews DE	Fine-tuned	63.6	3.5	68.0	4	0
10kGNAD German news	LogReg	84.1	6.6	89.8	55	26
10kGNAD German news	Fine-tuned	88.6	6.5	95.3	84	41
Swiss Federal Supreme Court decisions DE	LogReg	89.0	3.0	97.0	86	64
MASSIVE intents DE	LogReg	87.0	3.6	94.0	43	0
MASSIVE intents DE	Fine-tuned	87.0	3.6	94.0	40	0
MTOP intents DE	LogReg	94.5	2.1	98.7	99	66
MTOP intents DE	Fine-tuned	94.3	3.9	98.8	99	64
Open Food Facts DE	LogReg	44.9	3.8	52.7	2	0
AG News	LogReg	87.2	2.6	93.7	71	7
Medical abstracts	LogReg	70.5	6.0	76.6	12	0

Dataset	Classifier	Accuracy	ECE	Accuracy on the 80 % most confident	Automatable at 95 % accuracy	at 99 %
Emotion	LogReg	92.4	9.0	98.5	93	71
Yahoo! Answers Topics	LogReg	67.1	5.7	75.2	11	1
Yahoo! Answers Topics	Fine-tuned	71.5	5.8	80.0	21	1
arXiv papers	LogReg	82.9	6.3	90.3	61	16
arXiv papers	Fine-tuned	86.6	5.8	93.7	76	51
DBpedia-14	LogReg	98.2	1.1	99.6	100	99
20 Newsgroups	LogReg	65.7	9.7	75.3	23	2
20 Newsgroups	Fine-tuned	72.7	13.0	83.0	44	14
Banking77	LogReg	93.0	1.3	98.8	96	72
Banking77	Fine-tuned	92.6	4.8	98.6	95	73
LEDGAR contract clauses	LogReg	74.2	15.8	82.2	41	12
CLINC150	LogReg	92.6	6.0	98.6	95	76
CLINC150	Fine-tuned	95.2	2.1	99.4	100	86
HUPD patents	LogReg	50.2	8.4	56.7	0	0

- LogReg and the fine-tuned encoder are reasonably calibrated (ECE mostly 1–7 %); their confidence can be thresholded directly.
- Fine-tuning increases the share of texts that can be automated even where overall accuracy rises little (arXiv 61 → 76 %, 10kGNAD 55 → 84 %, 20NG 23 → 44 % at 95 % accuracy).
- Ensembles with TF-IDF output softmaxed SVM scores: high accuracy but poorly calibrated confidence (Banking77 ECE 41 % for the char ensemble, 68 % for fine-tuned + TF-IDF). Recalibrate them (e.g. temperature scaling on validation) before using confidence thresholds.
- On many-class and subjective tasks (HUPD, Open Food Facts, Amazon stars, GermEval) no threshold reaches 95 % accuracy on more than a few percent of texts.

Input length and embedding size



The longer window helps only in proportion to how many texts exceed 512 tokens. Matryoshka truncation: 512 dimensions −1.1 points on average (worst −3.1), 256 −2.8, 128 −5.2. The loss below

256 dimensions varies widely between datasets (see the figure).

Cost

Embedding runs at about 11,000 tokens/s and LoRA fine-tuning at about 1,700 tokens/s on the shared RTX 3060; both scale with tokens, not with the number of classes. LogReg, kNN and TF-IDF train in seconds to minutes on CPU, except TF-IDF with character n-grams on large many-class sets (HUPD: more than 11 GB of RAM, skipped).

Recommendations

Default pipeline. Embed once, train LogReg (tune C on a validation split), and add a TF-IDF SVM in a probability ensemble. That is the best or within a point of the best on most datasets, costs one embedding pass, and retrains in minutes when labels change.

Larger datasets (≫ 10k training texts). - Embed once and keep the embeddings: classifier experiments then cost CPU minutes, not GPU hours. - Fit LogReg with a fast solver (SAGA or a GPU softmax layer) and a coarse C grid; subsample for SVMs and for hyper-parameter search (we used 30k for the RBF SVM). - An MLP becomes worth trying above roughly 20k training texts (LEDGAR +2.6 points), still with a 1-layer default. - Fine-tune only on a capped subset first (e.g. 1,000 per class): the learning curves show diminishing returns, and fine-tuning time grows linearly with tokens (about 1,700 tokens/s on an RTX 3060 for LoRA on jina v5 small). - Watch memory for TF-IDF on many classes: character n-grams on 98k texts × 408 classes needed more than 11 GB of RAM; use hashing with fewer features or word n-grams only. - Use 512-d Matryoshka embeddings when storage or kNN search matters; budget about 1 point of accuracy.

Many classes (> 100–150). - Do not rely on frozen embeddings alone: always include TF-IDF (word and character n-grams) and the ensemble; on 342 and 408 classes it was 4–16 points ahead of LogReg. - Report balanced accuracy, macro-F1 and top-3 alongside accuracy; macro-F1 on HUPD is 16 points below accuracy. - Make sure each class has enough examples (≥ 20, ideally ≥ 100); merge or drop catch-all and near-duplicate categories (as done for Open Food Facts). - Consider a hierarchy: predict the parent category first (e.g. the IPC class), then the subclass among its children. - Use class weighting, and check tail recall. - Zero-shot and kNN are not viable at this scale.

Short texts with semantic classes (intents, emotions). Frozen embeddings + LogReg are enough (87–95 %); kNN is a good alternative when classes change often, because no retraining is needed. Fine-tuning brings little.

Long documents. Use the 2,048-token window (or head + tail when fine-tuning) and combine with TF-IDF; legal and scientific texts carry much of their signal in vocabulary.

German. jina v5 handles German well on frozen embeddings and is the practical choice when one embedding store serves several tasks. When one task justifies fine-tuning, train `gbert-large`: it was 1–3 points ahead of LoRA-fine-tuned jina on all three German datasets, at lower training cost.

When to fine-tune at all. Fine-tune when classes are topical and the frozen + TF-IDF ensemble plateaus; then ensemble the fine-tuned model with TF-IDF again. Skip it for short intent and emotion texts, where the frozen encoder is already as good. After fine-tuning, keep the linear head; re-running the classifier sweep on the fine-tuned embeddings brought nothing.

Routing to manual review. Use a calibrated classifier (LogReg or the fine-tuned model, not the raw ensemble scores) and pick the confidence threshold on validation for the accuracy you need; the reliability tables show how much can then be automated. Check significance before switching methods: on 11 of 18 datasets the top methods are statistically tied.

Imbalanced data. Decide first whether accuracy or rare-class recall matters. Class-weighted LogReg costs nothing to try; logit adjustment can be tuned after training on validation; with hundreds of classes expect to trade several points of accuracy for large tail-recall gains.

Still running

- LoRA fine-tuning, fine-tuned + TF-IDF ensembles and the fine-tuned-embedding sweep on Emotion, DBpedia, AG News, Open Food Facts, Medical abstracts, LEDGAR and HUPD (1 epoch).
- gbert-large on GermEval 2018, MTOP, Open Food Facts and Swiss decisions.
- Plain, class-weighted and logit-adjusted fine-tuning on Swiss decisions (repeated after an out-of-memory failure), character TF-IDF on HUPD (smaller feature space).

Reproduce

Code, data preparation and all raw results are in this repository: `results/results.json` (every number above, incl. per-setting validation scores), `bench/` (pipeline), `bench/report.py` (these figures and tables), `docs/` (the interactive dashboard, also on GitHub Pages). See the README for setup and the GPU server configuration.